

Modeling Epistemic Vigilance in Collaborative Problem Solving: A Multimodal Approach Using Social Deduction as a Proxy Testbed

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Abstract

AI agents that monitor collaborative learning must detect reasoning failures in real time: misconceptions spreading unchecked, weak claims accepted without scrutiny, or participants reasoning actively but reaching wrong conclusions. Building detection models requires datasets where these failures actually occur and are labeled and yet existing collaborative datasets capture only idealized cooperation under shared epistemic goals. Detecting these failures also requires participant-relative signals such as gaze direction, attention targets, and social cues from each person’s viewpoint that exocentric sensing cannot reliably recover. We present a testbed addressing both gaps, combining a three-layer annotation framework (communicative acts, player-reported belief states yielding first- and second-order Theory of Mind data, and post-hoc epistemic vigilance assessments) with multi-person egocentric capture via Meta Aria research glasses alongside exocentric recording. Using a social deduction game as a structured environment where unreliable information enters group reasoning with known provenance, we instrument every participant, yielding synchronized first- and third-person streams. Pilot data demonstrates that failed vigilance, belief fragmentation, and traceable misinformation effects emerge naturally and are captured by the framework.

1. Introduction

An AI agent embedded in a collaborative classroom should be able to detect when collaboration is going wrong. When a student confidently asserts an incorrect solution, the agent must recognize the claim as unsupported. When the group accepts that claim without scrutiny because of social dynamics, groupthink, or simply because the loudest voice carries the most weight, the agent must detect that critical evaluation has failed. When argumentation devolves from evidence-based reasoning into a dominant participant steamrolling the group’s shared understanding, the

agent must distinguish counterproductive influence from constructive leadership. These capabilities of detecting misinformation, identifying poor reasoning, flagging absent evidence, and recognizing when group dynamics have overtaken evidential merit are what teachers do when they monitor group work, and they are what an AI agent needs to provide meaningful support [3, 4].

Building models with these capabilities requires training data where the failures actually occur and are labeled. This is a core problem: existing collaborative datasets capture idealized cooperation under shared epistemic goals [9, 15], where dynamics such as misconceptions propagating through groups, weak arguments accepted uncritically, overconfident errors going unchallenged, social loafing are not surfaced. Without data where epistemic failures occur with known ground truth where we know who introduced the misleading claim, who accepted it, and whether that acceptance was warranted, building detection models is intractable.

A second gap concerns sensing. Existing multimodal learning analytics rely on exocentric sensing: ceiling-mounted cameras and external microphones [1]. These may not reliably recover participant-relative signals: whether someone is looking *at you*, directing a statement *toward you*, or attending to the current speaker. Egocentric platforms exist—Ego4D [7], Ego-Exo4D [8], Project Aria [6] but their application has not been explored in a collaborative setting. We address both gaps with a testbed pairing multi-person egocentric sensing with a structured environment that produces labeled epistemic failures. Using Secret Hitler, a social deduction game where unreliable information enters group reasoning with known provenance, we instrument every player with Meta Aria research glasses alongside an overhead camera. A three-layer annotation framework traces communicative acts through belief updates to epistemic vigilance assessments, providing ground-truth labels for the reasoning failures described above alongside the multimodal signals needed to detect them.

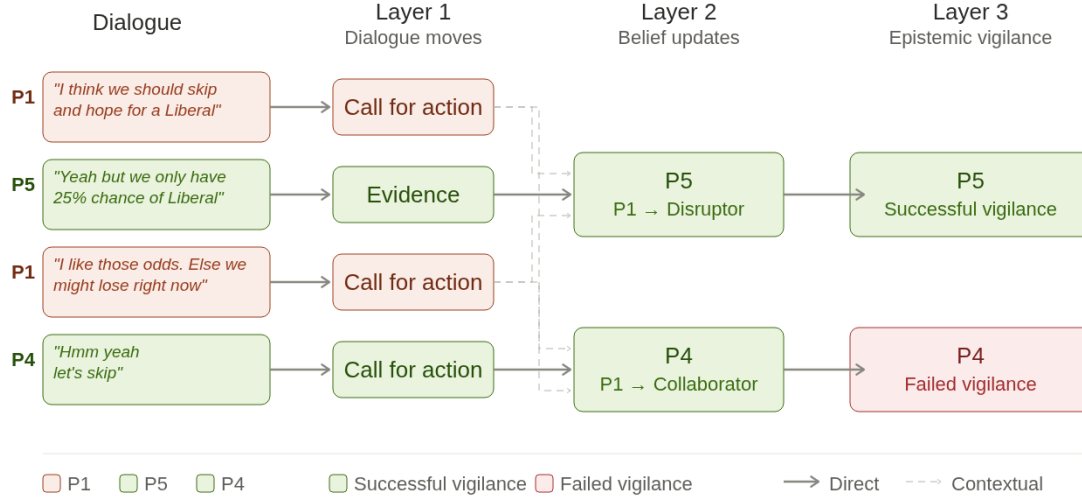


Figure 1. Three-layer annotation of a discussion phase. Utterances are classified by communicative function (Layer 1), linked to belief updates (Layer 2), and assessed against ground truth as successful or failed epistemic vigilance (Layer 3). Color indicates true role (red = adversarial, green = collaborative; hidden during play). P1 advocates for an irrational move of skipping a turn and hoping for a favorable outcome. P5 correctly objects by presenting evidence that the odds are poor, exhibiting successful vigilance. P4 accepts P1’s reasoning and agrees to skip, showcasing failed vigilance.

2. Testbed Design

2.1. Environment and Ground Truth

In Secret Hitler, 5 to 10 players are secretly assigned roles as Liberals or Fascists. Each round follows a structured sequence: a President nominates a Chancellor, the group discusses and votes, and the approved government enacts a policy from a constrained deck. For Liberal players, this is collaborative problem solving under genuine uncertainty: they must pool incomplete evidence, evaluate the reliability of others’ claims, and coordinate decisions without knowing who is trustworthy [12, 15]. The game enforces the information asymmetry that the hidden profile literature identifies as the hardest aspect of real collaboration [11, 14]—a challenge most CPS testbeds sidestep by giving participants shared access to the same information.

Fascist players are structurally incentivized to introduce unreliable information: asserting false claims with confidence, selectively withholding evidence, and undermining group reasoning. These contributions are functionally equivalent to the misconceptions, overconfident errors, and unproductive dominance documented collaboration [4]. The collaborative challenge they pose is the same, even though the underlying intent differs. Crucially, every piece of unreliable information has known provenance (role assignments provide ground truth), enabling traceable study of how epistemic failures propagate. We do not claim game players are cognitively equivalent to struggling students; we claim the *epistemic challenge* of evaluating contributions of uncertain reliability under incomplete information

is shared.

2.2. Annotation Framework

The framework captures progressively deeper social-cognitive constructs, operationalizing Baker et al.’s [2] Bayesian Theory of Mind and Sperber et al.’s [13] epistemic vigilance for multimodal behavioral annotation.

Layer 1: Communicative Acts. We adopt the six-category persuasion strategy taxonomy from Lai et al. [10]: Identity Declaration, Accusation, Interrogation, Call for Action, Defense, and Evidence, annotated at the utterance level (Krippendorff’s $\alpha > 0.6$). This captures what each speaker is doing communicatively.

Layer 2: Player-Reported Belief States. After each round, players report (a) *first-order* beliefs about every other player’s role (Liberal, Fascist, or uncertain), yielding calibration data against ground truth, and (b) *second-order* beliefs about what others believe about them, enabling direct measurement of Theory of Mind accuracy [2]. These reports provide supervision for models that must infer beliefs from observable behavior, since a classroom agent cannot ask students to report their beliefs directly.

Layer 3: Epistemic Vigilance. Inferred post-hoc using Layers 1 and 2 with ground-truth role assignments [13]. For each communicative act targeting a player, we assess whether the receiver’s belief update was warranted: *successful vigilance* (belief moved toward truth) or *failed vigilance* (belief moved away from truth—e.g., persuaded by a misleading claim, or failing to update when evidence warranted it). The cascading structure of communicative act



(a) Exocentric (overhead) view



(b) Egocentric (Aria glasses) view

Figure 2. **Exocentric vs. egocentric views from a pilot session.** (a) The overhead camera captures the full group, but one player on the left is partially occluded. (b) The egocentric view from one player’s Aria glasses reveals that the player’s attention is directed at a player across the table (center, hands raised), *not* at the adjacent player who is actively speaking.

→ belief update → vigilance assessment mirrors what a collaborative agent must track: a student proposes an incorrect solution (communicative act), peers accept it (belief update), and the agent detects that critical evaluation failed (vigilance assessment).

3. Multi-Person Egocentric Sensing

The annotation framework defines *what* to detect; the sensing infrastructure determines *what signals are available* for detection.

3.1. Sensing Setup

All n players wear Meta Aria Gen 1 research glasses [6], each capturing egocentric RGB video, eye-tracking data (gaze direction and pupillometry), and IMU signals (head orientation and movement). Group audio is recorded via the glasses’ built-in microphones. An overhead camera provides a synchronized exocentric reference view, and game state is logged by a facilitator, producing $n + 1$ synchronized video streams per session alongside continuous eye-tracking and inertial data.

3.2. Why Egocentric Sensing

Most multimodal learning analytics rely on exocentric cameras [1]. While useful for group-level views, overhead or wall-mounted cameras cannot reliably recover the social-cognitive constructs our framework targets: they suffer from mutual occlusion between adjacent participants, cannot distinguish fine-grained attention targets (speaker vs. another player’s reaction vs. game board), and miss participant-relative social cues such as directed eye contact or gaze toward an ally during an accusation which are signals available only from the receiver’s perspective and directly relevant to Layer 3 (see Section 4).

3.3. Gaze-Derived Features

The multi-person egocentric configuration enables three categories of gaze-derived features aligned with our anno-

tation layers:

Gaze target identity (Layer 1): Projecting each player’s eye-tracking ray onto their egocentric RGB stream and cross-referencing with exocentric detections identifies who each participant attends to at utterance onset. This answers: *was the receiver looking at the speaker during an accusation?*

Scanpath structure (Layer 3): Temporal gaze-target sequences can be analyzed using scanpath similarity metrics [5] to investigate whether players exhibiting successful vigilance show systematically different gaze patterns—e.g., checking multiple players’ reactions before updating—compared to those exhibiting failed vigilance.

Group-level attention divergence (cross-layer): Because every player is instrumented, we compute the entropy of the gaze-target distribution across all players at each timestep—a measure unavailable in single-viewpoint datasets [7, 8]. High entropy indicates fragmented attention; low entropy indicates convergence.

4. Pilot Observations

We report observations from a pilot session (6 players, 9 rounds, 2 Disruptive / 4 Collaborative). All players wore Aria glasses and completed belief reports after each round.

Failed epistemic vigilance. Figure 3(a) tracks collaborative players’ evolving beliefs about P5 (true role: Disruptive). Despite 9 rounds of discussion and evidence, no collaborator correctly identified P5 constituting failed vigilance: P3 possessed a partially correct signal but abandoned it, likely in response to P5’s communicative acts (Layer 1). Inspection of P3’s egocentric stream during the Round 3–4 transition reveals that P3’s gaze was directed away from P5 during P5’s defense, suggesting the first-person record may contain cues about *why* vigilance failed that the overhead view cannot provide.

Active but counterproductive participation. Figure 3(b) reveals that P3 correctly identified zero allies across all 9 rounds, placing every collaborator in the Disruptive

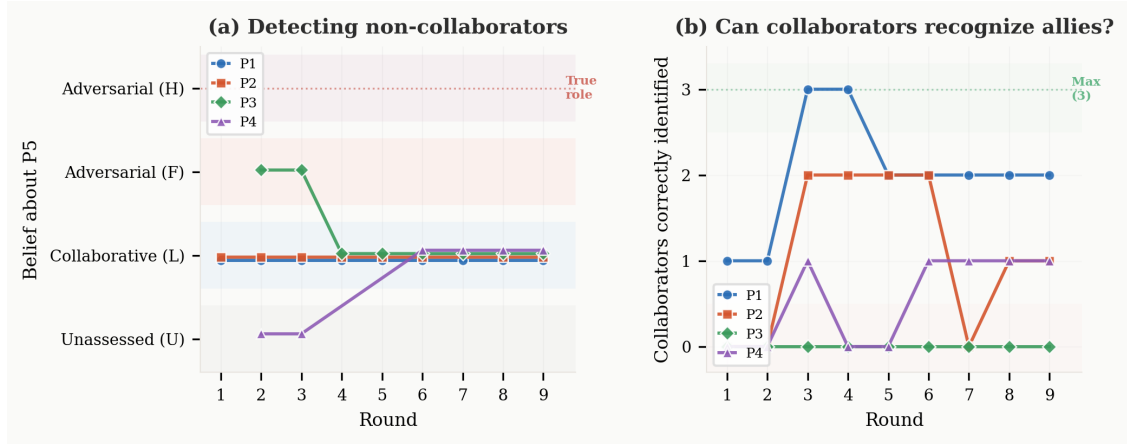


Figure 3. (a) Four collaborative players’ beliefs about P5 (true role: Disruptive) across rounds. No collaborator ever correctly classified P5, illustrating failed vigilance. (b) Number of fellow collaborators each player correctly identified per round (max = 3).

category—the profile of a group member who is actively reasoning but systematically misdirected. P1, by contrast, peaked at identifying all three allies (Rounds 3–4) before declining, suggesting group-level belief fragmentation as unreliable information accumulated.

5. Discussion

Monitoring group reasoning under adversarial pressure. Existing collaborative frameworks focus predominantly on productive cooperation on how groups build shared understanding when participants share epistemic goals. How groups *fail*, how misinformation propagates, how unsupported claims survive scrutiny, and how individual reasoning errors compound into collective ones, is far less studied, in part because such failures are sparse and unlabeled in naturalistic data. Our testbed addresses this directly: the game’s adversarial structure provides scaffolding that reliably produces these failure modes, while the annotation framework labels them with known provenance. This reframes the monitoring problem from detecting rare, ambiguous breakdowns in cooperative settings to studying frequent, ground-truthed failures in a structured environment, offering a complementary development path for classroom agents that must ultimately handle both.

What the testbed enables. The multi-layer framework produces *linked annotations*: specific communicative acts traced to specific belief changes in specific individuals, assessed for epistemic warrant, a cascading structure not available in existing CPS [15] or deliberation [9] datasets. The multi-person egocentric configuration captures participant-relative signals such as attention targets, visual perspective during key exchanges, and mutual gaze which are directly relevant to explaining *why* vigilance succeeded or failed. Together, these enable training models

that predict epistemic vigilance outcomes from multimodal cues, and comparing egocentric versus exocentric features for this prediction. We frame the testbed as accelerating model development impractical to iterate on in instructional settings, not as a direct classroom analog: the game magnifies the epistemic challenge, but the perceptual and inferential demands on the monitoring agent are shared with real classrooms.

Next steps. With additional sessions we plan to train classifiers that predict Layer 1, 2, and 3 labels from multimodal features, systematically comparing egocentric-only, exocentric-only, and fused feature sets. A key question is whether gaze-derived features, particularly scanpath structure and attention divergence, improve prediction over audio-visual baselines alone. We also intend to evaluate cross-session generalization and to explore whether models trained on game interactions transfer to collaborative learning tasks with shared epistemic goals.

Limitations. The pilot is a single 6-player session; we do not claim generalizability. Player-reported beliefs are subject to social desirability. Layer 3 depends on binary ground truth cleaner than classrooms provide. The gaze-derived features in are capabilities the configuration enables; systematic analysis of their informativeness is ongoing. The sensing requires research-grade hardware; the testbed’s value is in model development, not deployment.

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